

Unit -3 E-Governance Infrastructure and Strategies [8 Hrs.]

While e-governance is a reality beyond hype in developing countries, a fact usually not understood is that many e-governance initiatives fail before they are well entrenched on a permanent and sustainable basis.

With the proper infrastructure and strategies we need to answer following questions:

- **What are the reasons behind high mortality rate of e-governance?**
- **What are the pitfalls and bottlenecks?**
- **What is the solution for this?**

E-readiness

- **E-Readiness refers to a country's ability to take advantage of the Internet as an engine of economic growth and human development.**
- **E-readiness refers to a country's capacity and state of preparedness to participate in the electronic world.** The state of maturity is commonly measured by the country's information and communications technology (ICT) infrastructure and the ability of its government and citizens to use the positive impacts of ICT for sustainable development.
- **E-Readiness is the public sector's generic capacity or aptitude to use ICT to encapsulate public services and deploy to the public high-quality information (explicit knowledge) and effective communication tools that support human development.**

Developing countries face serious constraints and limitations or challenges to fully adopt e-governance. All these are related to the e-readiness of the country i.e. its readiness for e-governance implementation.

E-readiness means the infrastructural pre-requisites for taking up any e-governance initiative.

These infrastructural prerequisites or preconditions may be identified as:

- Data System Infrastructure Preparedness
- Legal Infrastructural Preparedness
- Institutional Infrastructural Preparedness
- Human Infrastructural Preparedness
- Technological Infrastructural Preparedness
- Leadership and strategic planning

Data System Infrastructure Preparedness

- The core of e-governance is e-MIS and holds the entire database of any organization.
- The major question that arises here is "Are all the requisite management information systems, records, databases, and work processes in the proper place so as to provide the quantity and quality of data to support the move to e-governance?"
- The data that were managed manually need to be computerized or brought into the electronic form which means that the preparedness of a computerized database or the data warehouse is required.
- Data quality and data security are of prime concern here as most of the government infrastructures are not up to the mark in developing countries,
- This is the core computerization activity of any government process which may take several years to reach this stage.

Legal Infrastructure Preparedness

- They lack requisite legislation and legal infrastructure to enable such reforms or reengineering of the existing business practices, rules, and regulations within the government at various levels.
- The manual processes in government are usually obsolete, inefficient, and bureaucratic.
- Though they have transformed to computerization practices, they continue to have poor and inefficient performance and this is due to a lack of administrative reforms and lack of business process reengineering.
- This seems to be accentuated in developing countries while developed countries have been significantly successful in administrative reforms and business reengineering.
- The fundamental question that arises here is" Are the laws and regulations required to permit and support the move towards e-governance initiatives in place? E.g Digital Signature Act

The efforts put in computerization, as described above, result only in automation of the existing manual processes and procedures. The manual processes and procedures in government are usually obsolete, inefficient, bureaucratic and red tape based. Some developing countries still continue their obsolete colonial practices in government. This means that by computerizing the same existing inefficient manual practices and systems we will continue to have poor and inefficient performance, though at a higher speed, by computerization. It is also due to lack of administrative reforms and lack of business process reengineering. It is also due to lack of requisite legislation and legal infrastructure to enable such reforms or reengineering of the existing inefficient, or obsolete business practices and procedure or rules and regulations within the government at various levels. This problem is more accentuated in the developing countries, some of which still have their colonial hangovers. In the developed countries, especially the smaller ones, administrative reforms and reengineering have been significantly successful. In the developing countries too, this has begun to some extent and the future holds hope for this as to enable effective e-governance implementation in these countries.

The fundamental question that needs to be answered in the affirmative before any successful e-governance institution is taken up is: Are the laws and regulations required to permit and support the move towards e-governance initiatives in place? For example, digital signatures are not yet accepted in many countries which lack the appropriate law for this. The Information Technology Act (IT Act) in India, essentially provides an appropriate legal infrastructure for the use of digital signature. The public key infrastructure is being given the legal sanction and support before it can be implemented. Apart from these statutory laws pertaining to IT, individual legal acts and legislation may be required for enabling smooth and successful implementation of specific e-governance initiatives (e.g., the legal acceptance of a computer printed document). Many countries have legislated the IT Act (as in India) which needs all these requirements.

Institutional Infrastructural Preparedness

- For any government to implement a successful e-governance project, the required institutional infrastructure must be in place which most of the government lacks.
- The government body has to establish a separate IT department that basically coordinates with facilitators for e-government projects within the nation.
- The IT department works out for the hardware selection and procurement, network or software development and implementation, and staff training at various levels of the government.
- Many countries still lack the institutional infrastructure.

For any government to implement successful e-governance projects, the requisite institutional infrastructure has to be in place. Most governments do not have any. For example, in India the Government of India established National Informatics Centre (NIC) as an apex institution at the national level for catalyzing and coordinating all e-government activities and projects in any government body at the Central, State and District levels. Similarly, many State governments in India have established their own Information Technology Departments which are basically coordinating facilitators for e-government projects within the State.

However, many countries or states still lack this institutional infrastructure. The activities of such institutes are varied from hardware selection and procurement to networking or software development and implementation and also training of the staff at various levels of the government.

Human Infrastructural Preparedness

- Human resource development by training is an essential requirement that comes from well-trained manpower both technical and non-technical.
- The technical manpower resources are essential for all the phases of e-governance and related information system life cycle comprising systems analysis, design, programming, implementation, operation, and documentation.
- Both private and government institutions should play a major role in this regard.
- Apart from technical human infrastructure, there is a need for the crucial training and orientation of user personnel i.e. government staff in e-governance projects.
- The government employees and staff who are the stakeholders in all e-government projects as the end-users are to be appropriately trained and oriented for change management from a manual government environment to an e-governance environment.
- Such training will make them competent and capable of handling e-governance projects at the operational level.

An institutional infrastructure provides training facilities in addition to other resources. Human resource development by training is an essential requirement. Human infrastructural preparedness comes from well-trained manpower, both technical and non-technical. The technical manpower resources are essential for all the phases of e-governance and related information system life cycle comprising systems analysis, systems design, programming, implementation, operationalization, and documentation. The national public IT infrastructural institutions (such as National Informatics Centre) for e-governance provide manpower for all such activities. Besides, the private corporations also play a major role in this regard in many e-governance activities. In addition to the requisite technical human infrastructure for software development and implementation of e-government projects, there is need for the crucial training and orientation of user personnel, i.e. government staff in e-governance projects. The government employees and staff who are the stake-holders in all e-government projects as the end users and operational users of such projects, are required to be appropriately trained and oriented for change management from a manual government environment to e-governance environment. Only after such training will they be competent and capable of handling such e-governance projects and operational environments. They also play a crucial role in various phases of e-governance information system life cycle in the phase of systems analysis and operational usage.

Technological Infrastructural Preparedness

- Technology is fast changing in the ICT domain and there is rapid obsolescence of software as hardware which requires great financial support time and again.
- Government organizations encounter this situation especially as their procedures to procure hardware or software are very inefficient and slow.
- The technological infrastructure in developing countries including computing and telecommunication is absent. As a result software and hardware may be incompatible.
- The major reasons are
 - cost of technology
 - Adaptability
 - Obsolescence
- This is a serious limitation to e-governance implementation.

Technology is fast changing in ICT domain. Rapid obsolescence of hardware and software and the maintenance and support they require results in great financial demands from time to time. Government organizations encounter this situation especially as their procedures to procure hardware, software, etc. are highly inefficient and delayed.

In the case of developing and underdeveloped countries, the latest technological infrastructure including computing and telecommunication is conspicuous by its absence. As a result, software and hardware also may not be compatible or efficient. The reasons are too many—cost of technology, adaptability, obsolescence, so on and so forth. This is a serious limitation for e-governance implementation. Innovative solutions to solve the problem of funding are being explored by these countries. The new technology domains as fibre optics enable low cost Internet bandwidth. Some countries are also making plans for broadband Internet access.

Evolutionary Stages in E-Governance.

E-governance evolves gradually from the simplest levels to advanced levels. The evolution may not be the same in all cases. Several stages of evolutionary progress could be identified. The following stages of e-governance can definitely be perceived in any government department or government organization:

- Stage 1.* Use of e-mail and setting up of internal networking
- Stage 2.* Creation of Intranets infrastructure for access of internal activities
- Stage 3.* Allowing public access to information through Internet
- Stage 4.* Allowing two-way interactive communication with stakeholders to enable Internet enabled transactions (including financial transactions)
- Stage 5.* Allowing online transactions by the citizens
- Stage 6.* Enriching digital democracy
- Stage 7.* Electronically integrated or joined government with Legislature and Judiciary.

Stage 1: Use of e-mail and setting up of internal network

The most fundamental, cheapest and easiest ICT tool is e-mail. E-mail has now become common in all urban areas and some rural areas. Within the government it is the easiest of all options available for implementation. Official orders to accept e-mail communication as valid have been issued in a large number of government, judicial and legislative organizations. Although e-mail can reach outside organizations via the Internet, most government organizations adopt e-mail for internal messagings.

Due to its informality, e-mail can lead to increased lateral and bottom-up communication.

E-mails break the official hierarchy of communication, as anyone can send to e-mail to any other, breaking the hierarchy and other barriers. They allow person-to-person communication can improve information sharing, exchange, coordination and feedback of information.

However, its limitations are its transparency and security risks of the content, unless used with digital signatures or key encryption. Sensitive and critical messages are still best sent over the telephone or in person. However, with the advent of digital signatures and encryption techniques, it is possible to ensure integrity, security, correctness and non-repudiation of the information sent as e-mail.

The internal networking of various departments of an organization linked to Internet for sending and receiving the e-mail is a prerequisite. Most government organizations have already set up desktops in various divisions and sections and internally connected them all for this purpose.

Stage 2: Use of Internet by connecting internal activities to Internet

While e-mail provides the very fundamental mode of communication, the basic and personal use of Internet from offices and houses is now generally a reality in all urban areas and limited rural areas. Surfing of the web is both a business and pleasure. It has been noticed that most government employees spend a few hours a day surfing the Internet whether for official purpose or personal benefit, or for pleasure. The Internet has inculcated an information culture in the people in general and government employees in particular, to surf the Internet, in general, for all purposes of information retrieval. Thus, all sectors of the government such as Agriculture, Finance, Economy, Planning, Rural Development, etc. can be found to have their presence on the Internet in a significant scale and any government activity regarding any information such as policy statements, strategies, technological or scientific information can be obtained from the Internet for the benefit of all levels of people involved in government. For example, the Andhra Pradesh Chief Minister's Office has set up a group of technical and administrative staff only to research and retrieve information from the Internet from time to time to prepare reports and presentations. Similarly, the office has also set up an official web site giving all activities of the Chief Minister and summary of government activities.

Stage 3: Allowing public access to information

Public access to information can be viewed in multiple dimensions.

- (a) **Web pages, citizen charters and application forms.** If the government department concerned puts up a web page on the Internet, describing all functions of the Department, it can be accessed by citizens and general public interested. This may be usually static data—the Internet equivalent of a printed brochure. The basic profile and functions of the Department concerned will be on Public display.
- (b) **General information services.** In addition to this e-enabling of the basic functionality of the government departments concerned, there could be much greater scope, content and depth of information that can be of interest and use for the people, and the citizens at large, that is the general information requirements. The general information requirement of the public in general can be broadly defined to a very large extent or scope and coverage, in wide ranging sectors such as Education and Tourism, apart from information from Census and the statistical information pertaining to various sectors of economy such as agriculture, industrial activity, plan details, etc. Before the Internet was established and became popular, a pioneering initiative was taken by National Informatics Centre (NIC) in the form of the Project GISTNIC (General Information Service Terminal of National Informatics Centre), the goal of which was to collate, compile and provide a single-point source of information of various sectors of interest to public in general. A wide range of subjects as, for example, Census data of all villages—population abstracts as also details of amenities were provided in addition to information of all tourist spots, information regarding universities and colleges, etc. Subjects such as rural technologies and specialized subjects as traditional sciences and technologies were also covered.

Stage 4: Allowing two way interactive communication with stake-holders to enable Internet enabled transactions (including financial transactions)

Once a web site is operational, correspondences from the citizens can be allowed through e-mail by providing them the appropriate e-mail addresses. For example, in China, the Beijing city government web site provides e-mail section to citizens apart from other important information such as government regulations, rules, laws or information about services offered by the government. In this e-mail section, the citizens are asked to express their suggestions, ideas, complaints (if any). The appropriate office concerned will be receiving these e-mails. Another approach is to permit citizens to clarify some queries (such as how to move the residence to Beijing) and the response to such queries will be posted on the web site itself.

The two-way interaction need not necessarily be online and web based. In the case of CARD (Computer aided Administration of the Registration Department) Project (executed by the Government of Andhra Pradesh and NIC in Andhra Pradesh State), title deeds are registered in one hour and encumbrance certificates are issued in 20 minutes. The entire set-up is within the concerned sub-registrar office and managed by the office staff themselves. The citizens are provided the requisite services by the officers concerned in a very efficient manner.

Stage 5: Allowing online transactions by the citizens

In stage 4, the citizens interacted through kiosks, obtained services through online or the Internet, but made payments manually. But now, in Stage 5, in addition to permitting single online enquiry access to information, citizens may be enabled to make payments of fees and taxes, lodge complaints, file applications and perform any other transactions online through citizen kiosks installed at busy public locations. This is a much more advanced stage in e-governance not yet reached in developing countries but already reached in the developed world.

In Singapore, the citizens can transact every government business online and round the clock through specially designed kiosks which can be operated using smart cards. They can transact all government related business such as social welfare claims, tax assessment and payment, visa applications and license renewals, in addition to bank based fund or financial transactions using smart cards. However, this is a very advanced technological scenario, too advanced to be replicated immediately in all developing countries, even though plans are being drawn in developing countries to execute similar initiatives. Of course, limited transactions with the government by the citizens have been made online in these countries. The kiosks in public locations in Beijing, China, are successful; this is true of Korea as well as of India (for land record details).

Still, in these countries the financial transactions have not been made online, even though collections may be allowed alternatively (by an operator collecting money). The main reason for not executing financial transactions on the Internet has been the lack of security—lack of confident and secure financial transactions on the net. With the imminent implementation of public key infrastructure in the entire world, including the poor and developing countries, the confidence on financial transactions over the net may be developed, and with the proliferation of the smart cards, and online payments may be very soon a way of life in the entire developing world.

As regards the G2B (Government to Business) transactions, there have been impressive advances in various developing countries. For example, in the Philippines, the Customs Bureau has enabled electronic payments of customs duty, electronic processing of clearance of documents and release of shipments. The new online system has resulted in fast and secure transmission of payment details. The time for reconciling of payments collected by banks and remittances to the national treasury has been reduced to a few days from several months. An information system called “Selectivity” categorizes shipments into high, medium or low risk transactions so that they can be coursed through appropriate examination procedures. This reduces fraud, corruption and other undesirable effects that normally result from the personal interaction of the officials with the business customers of the customs department.

Similar efforts, even though limited, have been made in other countries such as India, Thailand and Korea. In Thailand, the Customs Department has eliminated all manual processing, thus increasing efficiency and transparency. Similar effort is in place in Indian Customs Department which has started implementing Electronic Data Interchange (EDI). In Korea, the Public Procurement Service, a central government organization responsible for procuring commodities and arranging contracts for construction projects, has computerized the purchase process and accounting transactions using the EDI.

Cybershopping or e-procurement a growing trend in all countries, is popular for its ease of operation, efficiency and transparency. Contracting also is largely getting computerized. Databases of supply firms information are being set-up. Also, pre-qualification and cost accounting procedures are also being automated.

In countries such as Singapore, complete Enterprise Resource Planning (ERP) implementation has resulted in making compulsory the supplying firms to be e-enabled and compatible to the ERP environment (such as SAP).

Internet based tendering and contracting process can be designed in such a manner that the documents from contractors including performance records can be obtained using computer networks of relevant organizations directly instead of receiving them from middle men manually. This prevents false documents and data being supplied.

In many of these efforts, there is some scope for the government offloading some of its activities to private contractors who will operate the requisite services and earn their own revenues from the customers who may be individuals or business houses. There are models as Build, Own and Operate (BOO) or Build, Own, Operate and Transfer (BOOT) models. There have been some success stories in this regard all over the world. With increasing public sector reforms and privatization, these models may be more attractive than the conventional government owned operations of services. For example, in Hong Kong, the government web portal is entirely financed and maintained by a private company, thereby reducing the cost and risk to the government. Other governments also are expected to follow the lead and involve private sector in creating partnerships with suppliers and customers together with whom they can find ways to cut costs, improve quality and share benefits. The private partnership of the government should be taken up with caution as some of the information in government may be confidential and sensitive sometimes (though not always); also, the profit motive of private companies may not be always achieved by the government in developing countries. It is preferable that the government uses its own agencies for such purposes.

Stage 6: Enriching digital democracy

Democracy can be strengthened and enriched by ICT in multiple ways and modes. At least two important sets of ICT applications that can potentially support participatory and democratic processes, specially in the developing countries have been identified. These relate to applications that enrich and further empower the civil society organizations, and enable the citizens to express themselves by voting in democratic processes through Internet or by any other electronic means.

Examples of success stories on both these two applications identified above can be cited in developing countries. In the Grameena Bank Project in Bangladesh, a cell phone hand set could be bought by a poor woman (with the help of bank loan) who would in turn rent it out to other poor men and women who work in the fields. Finally, this resulted in a commercially viable rural cell phone service leading to significant economic development.

The application of ICT in civil society groups and organizations is of a wide spectrum. Several groups can even combine forces, raise funds and even challenge multinational corporations. These challengers can be as diverse as NGOs, trade unions and other self-help groups. Their networking of people could be achieved through ICT applications.

ICT has been used by the governments in various countries for strengthening democracy, democratic processes and reforms of democratic processes.

It is expected, in the long term, digital democracy will come to some countries in another form: electronic voting. As in USA and in Japan, voting sites can be set up for Internet based voting in elections. However, this will be very different in developing countries. In India and also in the Phillipines, the Election Commission has used ICT in several ways: the voters' information were kept in its web site along with details of timings and location of polling booths. Dissemination of such information through Internet was enhanced with the help of News Media. However, in developing countries largely the voting process itself is either manual or partly electronic. In India and other developing countries, even though Electronic Voting Machines (EVMs) have been used for voting purposes, the entire operation is still manual, even though Internet was used for purposes other than voting itself. Voting on the Internet has now made a beginning in limited scale in many small polls, especially in the developed countries.

Stage 7: Electronically integrated or joined up government with Legislature and Judiciary

In the final stage of e-governance a comprehensive web portal and a smart card integrates information and services from various responsible government agencies. In this stage both horizontal integration of services across departments and vertical integration of service delivery is expected to take place. As already available in Singapore and also being experimented in many states such as Andhra Pradesh, on a web portal users can obtain services across different geographical levels of government within the same functional area and also access different functions. Thus, in a scenario like this, a citizen could submit a change of address on driving license and such a change would automatically be effected in all other sectors such as health, education, elections, taxation, etc. (thereby avoiding need for multiple filing). This is true of horizontal integration of services in an e-government. Citizens also can use portals to make payments and other transactions, obtain a checklist of enclosures required for an application, find answers to frequently asked questions (faq) and engage the services of relevant commercial enterprises. In Singapore and Hong Kong, such state-of-the-art portals are operational. The Government of Taipei (Taiwan) also has set up a "One Window" service on the Internet for tax administration, public health and e-commerce. Smart cards are gradually becoming functional in several Asian Countries for all such activities.

Stages of E-Government - according to UN, Gartner, and Layne & Lee study

Presence

Presence is the first stage of development and is the establishment of a placeholder for delivering information in the future. It represents the simplest and least expensive entrance into e-government, but it also offers the fewest options for citizens. **A typical example is a basic Web site that lists cursory information about an agency, such as hours of operation, mailing address, and/or phone numbers, but has no interactive capabilities. It is a passive presentation of general information.** Some observers refer to these types of sites as 'brochureware,' suggesting they are the electronic equivalent of a paper brochure.

Interaction

The second stage is interaction. **Although interactive Web-based initiatives offer enhanced capabilities, efforts in this group are still limited in their ability to streamline and automate government functions.** Interactions are relatively simple and generally revolve around information provision. These types of initiatives are designed to help the customer avoid a trip to an office or make a phone call by making commonly requested information and forms available around the clock. These resources may include instructions for obtaining services, downloadable forms to be printed and mailed back to an agency, or perhaps e-mail contact to respond to simple questions.

Transaction

The third stage in the evolution of e-government initiatives is a transaction. **These initiatives are more complex than simple information provision and embody the types of activities popularly associated with e-government. They enable clients to complete entire tasks electronically at any time of the day or night. These initiatives effectively create self-service operations for tasks such as license renewals, paying taxes and fees, and submitting bids for procurement contracts.** Although the level of interactivity is of a higher magnitude than second stage initiatives, the activities still involve a flow of information that is primarily one-way (either to the government or to the client, depending on the activity). The electronic responses are generally highly regularized and create predictable outcomes (e.g., approving a license renewal, creating a receipt, acknowledging a bid).

Transformation

The highest order of evolution for e-government initiatives is transformation. **Initiatives at this level utilize the full capabilities of the technology to transform how government functions are conceived, organized, and executed. Such initiatives would have the robust customer relationship management capabilities required to handle a full range of questions, problems, and needs.**

Currently, there are very few examples of this type of initiative, in part due to administrative, technical, and fiscal constraints. One of the distinctions of these initiatives is that they facilitate the seamless flow of information and collaborative decision-making between federal, state, local, public, and private partners. In other words, transformative e-government initiatives often seek to remove the organizational barriers that promote agency-centric solutions and, instead, promote customer-centric solutions. Some advocates suggest that, at its most advanced level, e-government could potentially re-organize, combine, and/or eliminate existing agencies and replace them with virtual organizations.

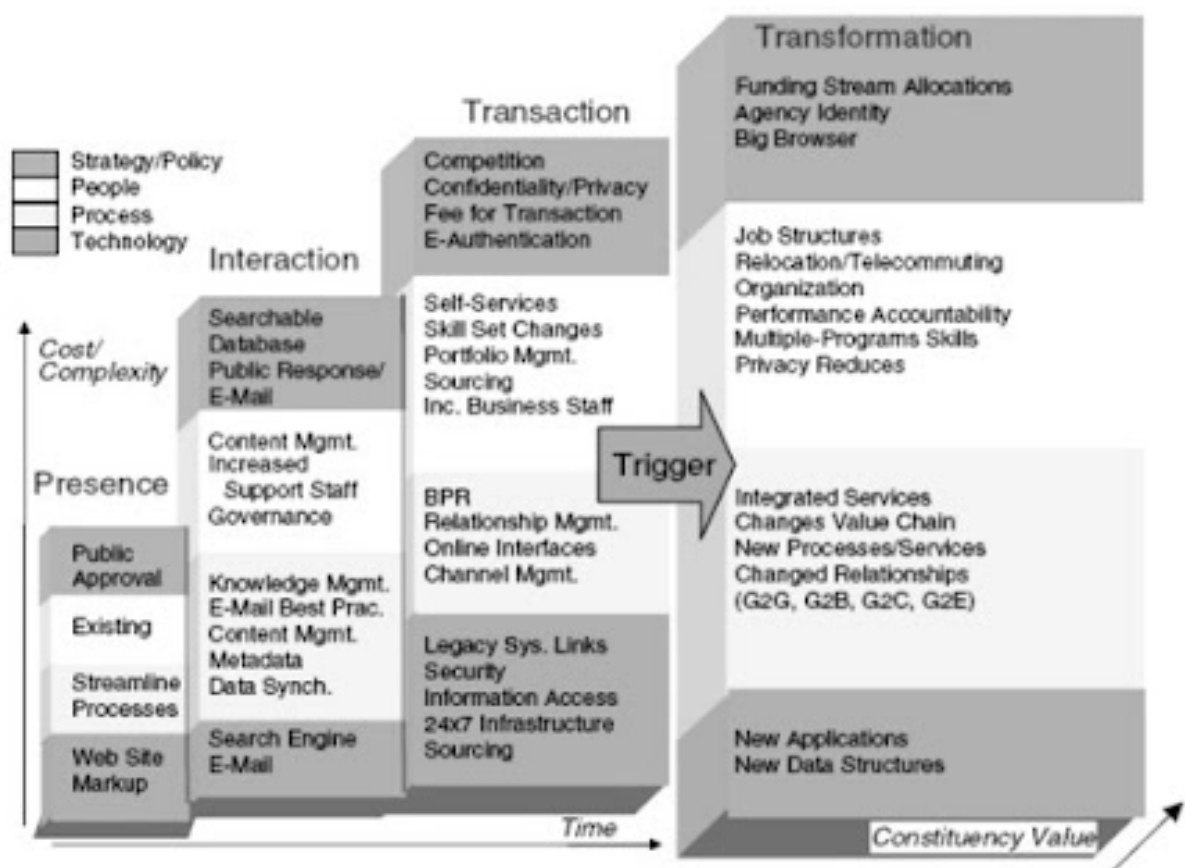
Gartner's Four phases of E-government model

Stage – 1 (Presence): Presence of simple information providing passive website indicating the same level information as information brochure.

Stage – 2 (Interaction): The interaction stage offers simple interactions between G2C, G2B, and G2G. This includes providing email contact, forms for taking in the questions, and providing informational responses.

Stage – 3 (Transaction): This stage involves the ability to make financial transactions to avail of government services

Stage – 4 (Transformation): Transformation stage involves making use of available data and learnings from transactions to transform governance and existing processes.



Four Phases of E-Government
Figure 1: Gartner Research, 2000

UN's Five stages of E-government model

Stage – 1 (Emerging): An online presence is established through the official website. Information is basic, static, and limited.

Stage – 2 (Enhanced): Government sites increase content, become more dynamic. The content is updated more frequently.

Stage – 3 (Interactive): Users can interact with the department through a set of forms and email.

Stage – 4 (Transactional): Users can actually avail themselves of government services by making payments on the website.

Stage – 5 (Seamless): There is a full integration of e-services across administrative and departmental boundaries.

Layne & Lee study – Four stages of e-government model

Stage – 1 (Cataloguing): Initial efforts of government is focused on establishing an online presence.

Stage – 2 (Transaction): In this stage, the department's website becomes able to transact electronically, and citizens can avail government services

Stage – 3 (Vertical Integration): Department attempts to integrate vertically with various levels of government such as local, state, and federal.

Stage – 4 (Horizontal Integration): Integration across different functions and services.

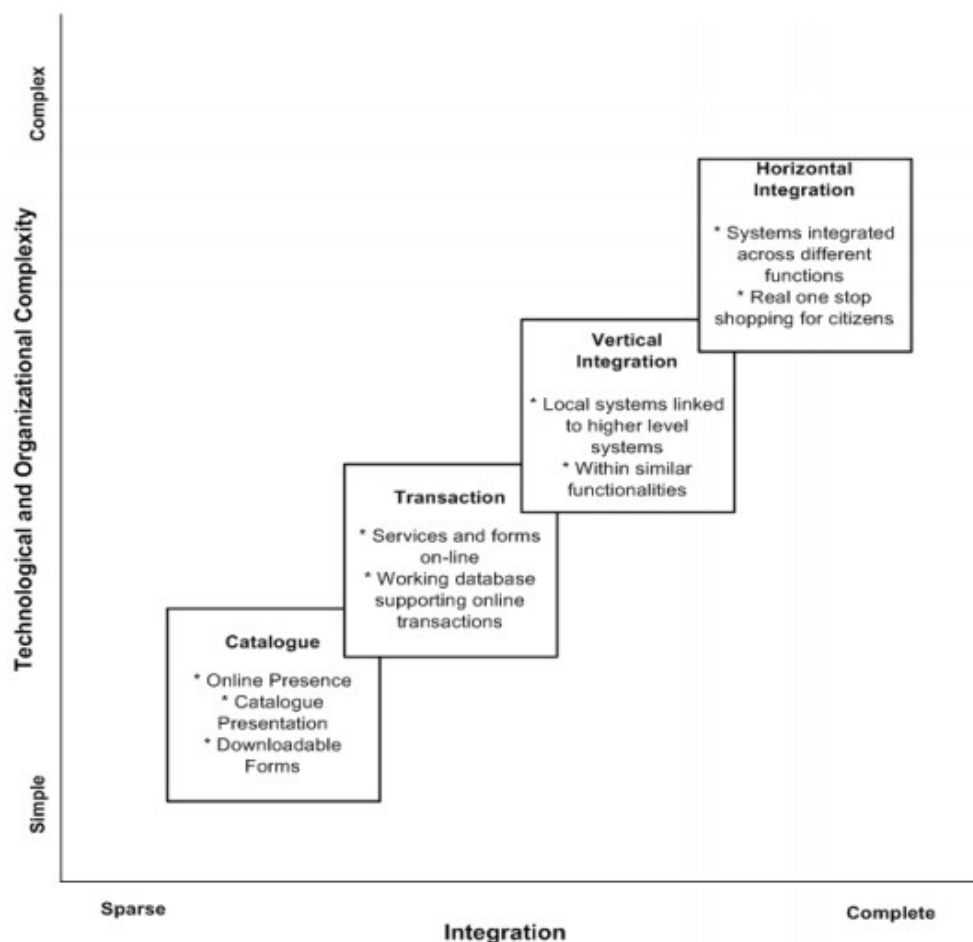


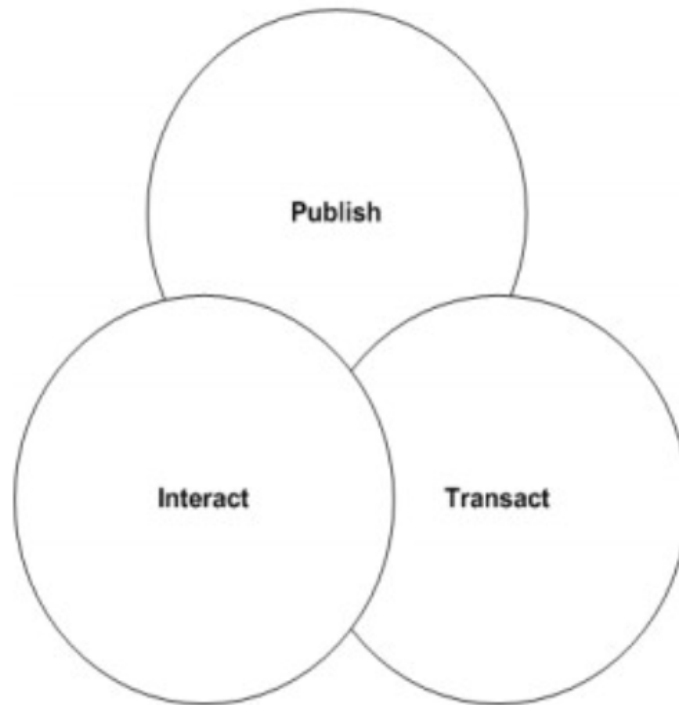
Figure 2: Layne & Lee, 2001 model

World Bank's three phases of e-Government model

Stage – 1 (Publishing): In this stage, government departments publish information on services, department related activities on their website.

Stage – 2 (Interact): Interaction with the government involves 2-way communication, where citizens can communicate and get a response back from the government through e-mail, submitting a form, or suggesting feedback.

Stage – 3 (Transact): In this stage, the government's e-governance efforts make the life of citizens easier by enables electronic transactions and simplifying government processes. Government, Business and Departments can avail services by making a payment.



Three Phases of E-Government

Figure 3: (Center for Democracy and Technology, 2002)